



Dead reckoning

Find your position without a compass or gps using the ancient technique of dead reckoning. <https://digit.in/jun20-68>



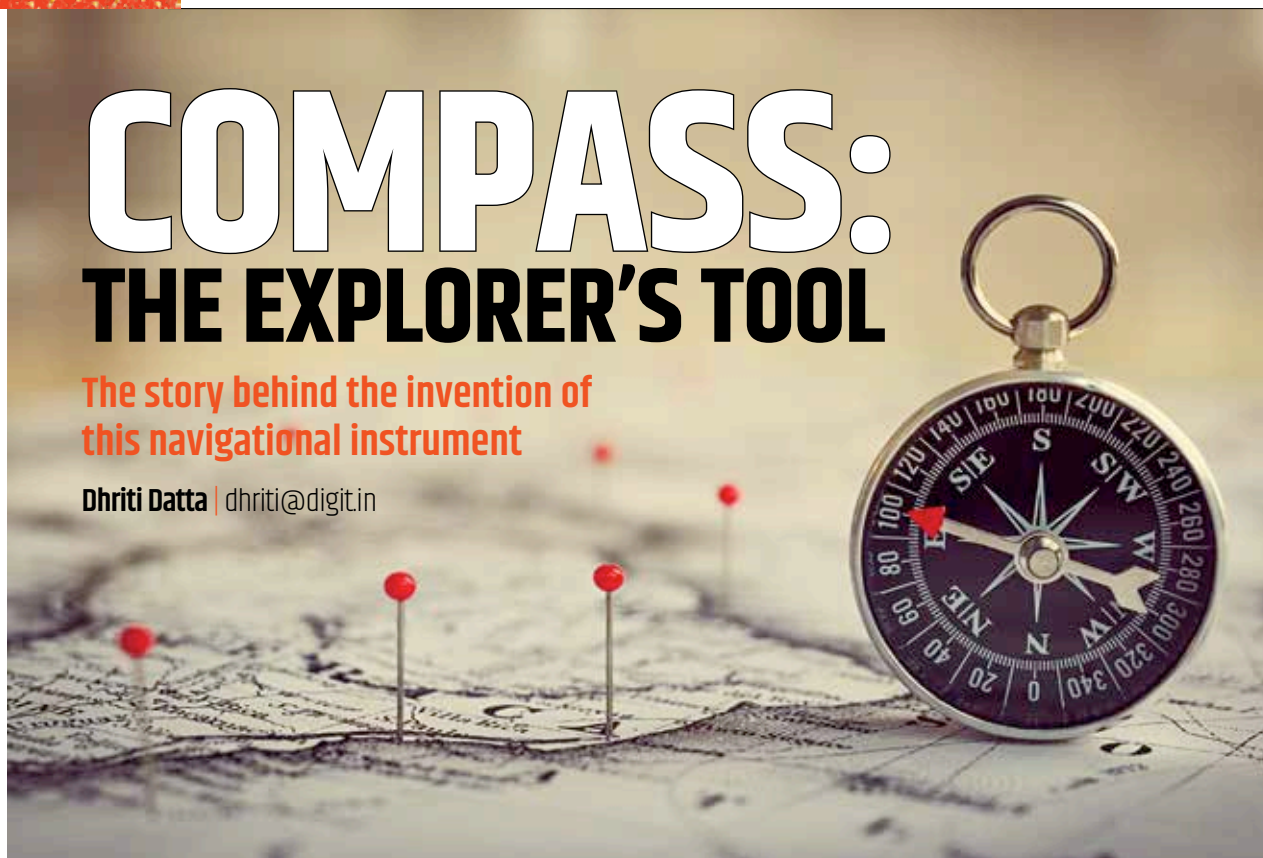
Biological compasses

Migratory birds have a specialised protein in their eyes that works like a magnetic compass. <https://digit.in/jun20-69>

COMPASS: THE EXPLORER'S TOOL

The story behind the invention of this navigational instrument

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nowing the right way to go was not always as effortless as it is today in our modern, technologically-fueled

world. We now have elaborate sensors and apps on our smartphones that work in tandem with GPS satellites and microwave towers to let us know our exact location, the routes to locations, and more. For ancient explorers, sailors and navigators, this proved to be quite the challenge. In ancient times, getting lost at sea or even on land was a tangible danger with consequences that could often be disastrous. Ancient Greek and Roman sailors even shortened the sailing season in the Mediterranean Sea due to precarious weather conditions which obstructed sight. In cloudy, overcast weather conditions, sailors and explorers couldn't even enlist the help of the sun and stars to help them navigate through treacherous waters or roads. All of this changed exponentially with the invention of the magnetic compass.

A compass is a navigational instrument which is used to determine

directional orientation. It is a device that features a magnetised needle that indicates the direction of the 'magnetic north' and bearings from it. The direction it shows is relative to the cardinal directions (also known as cardinal points) – North, South, East and West. With the invention of this navigational device, trade increased manifold since travel was made much safer and more efficient. It is touted to be one of the 'Four Great Inventions' from ancient China. Let's learn all about the history of the compass

THE VERY FIRST APPEARANCES

The compass first appeared between 300 and 200 BCE in China's Han Dynasty. The compasses were crafted out of lodestone, a naturally-magnetised ore of iron, which, if uninhibited by friction and gravity, would naturally align itself with Earth's magnetic field and point towards the magnetic poles. Ironically, these compasses made in China were not used for navigation at the time of their inception but for geomancy (a form of divination) and fortune-telling. They were utilised to

establish order and harmony to rooms or buildings in accordance with the geomantic principles of feng shui (famous Chinese geomancy). They were also used to choose suitable areas for building homes, growing crops and searching for gems.

The Chinese, during this time, began fashioning the lodestone into a shape that resembled a soup spoon or ladle. The lodestone spoon was placed at the centre of a bronze disk and the ore's magnetic properties made it spin around. The bronze disk had the cardinal direction points, direction indicators for constellations and other important symbols inscribed on it. When the lodestone stopped, the handle of the spoon would point south while the bowl part would look towards the north. These compasses were termed 'south governor' (sinán) or 'south pointers'.

USAGE AS NAVIGATIONAL DEVICES

The earliest reference of compasses being used as direction-finding devices for land navigation was during the Song Dynasty between the 9th and 11th